

PASSWORD MANAGER USING PYTHON

1. Introduction

The Password Manager is a simple Python-based application that helps users securely store and manage passwords for different websites. It allows users to save passwords, view stored passwords, and generate strong random passwords. The application uses file handling to store password data permanently in a text file.

2. Objective

- Store website passwords in a file.
- Retrieve saved passwords whenever needed.
- Generate secure random passwords.
- Demonstrate Python concepts such as dictionaries, file handling, loops, functions, and exception handling.

3. Software Requirements

Python 3.x, Text Editor (VS Code/PyCharm/IDLE), Windows/Linux/Mac OS.

4. Modules Used

Random Module and String Module for password generation.

5. Program Description

The application maintains a dictionary where website names are keys and passwords are values.

6. Working of the Program

1. Load existing passwords from passwords.txt using file handling.
2. Generate random passwords using letters, digits, and special characters.
3. Display a menu-driven interface.
4. Save passwords into dictionary and file.
5. View saved passwords.
6. Generate secure passwords.
7. Exit program.

7. Algorithm

1. Start program.
2. Load passwords from file.
3. Display menu.
4. Read user choice.
5. Save/View/Generate/Exit based on choice.
6. Stop.

8. Flowchart

START → Load Password File → Display Menu → Enter Choice → Save/View/Generate/Exit → END

9. Features

- Stores passwords permanently.
- Generates random passwords.
- Menu-driven interface.
- Uses file handling and exception handling.

10. Advantages

Simple, lightweight, easy to understand, and useful for organizing passwords.

11. Limitations

Passwords are stored in plain text and no encryption is used.

12. Future Enhancements

Add encryption, master password authentication, database storage, update/delete options, and GUI support.

13. Conclusion

The Password Manager project demonstrates important Python concepts including dictionaries, file handling, functions, loops, and exception handling. It serves as a good beginner-level project and can be enhanced with advanced security features.